ENGINEERING MATHEMATICS-II

Department of Science and Humanities

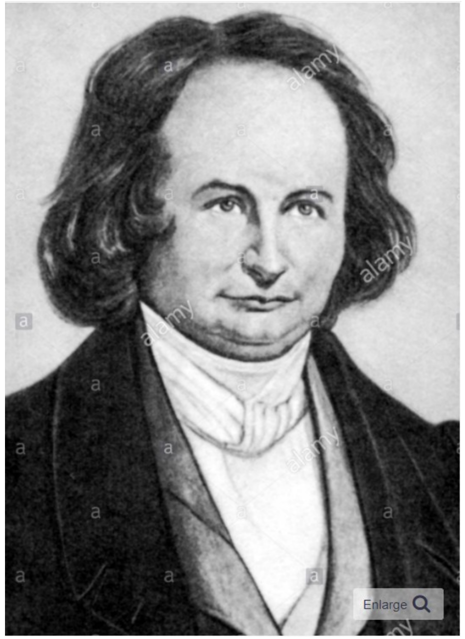
ENGINEERING MATHEMATICS-II
UE20MA151

UNIT 1: INTEGRAL CALCULUS
Class 3: jacobians

JACOBIAN MATRIX AND JACOBIAN

The Jacobian, named after the German mathematician

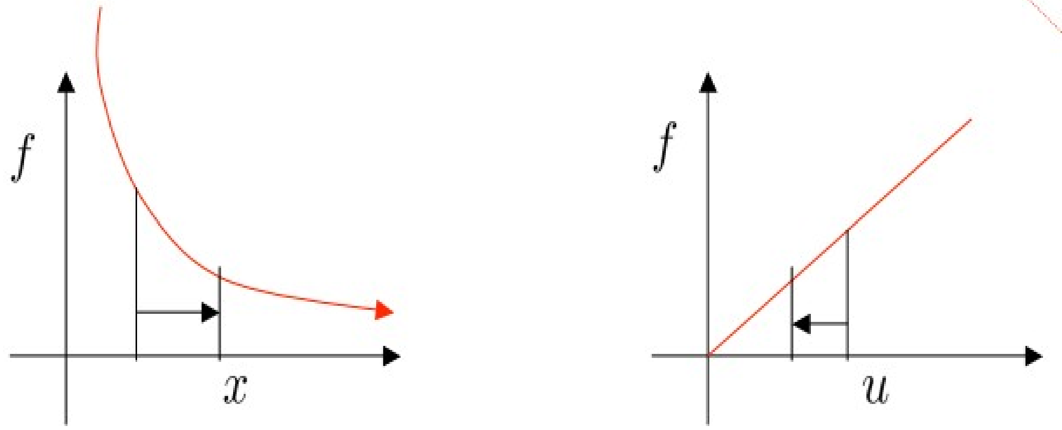
Carl Gustav Jacobi (1804-1851)



JACOBIAN

- In 1D problems we are used to a simple change of variables, e.g. from *x* to *u*

$$\int_a^b f(x)\mathrm{d}x = \int_\alpha^\beta f(x(u))\frac{\mathrm{d}x}{\mathrm{d}u}\mathrm{d}u$$



1D Jacobian

maps strips of width d*x* to strips of width d*u*

- Example: $\int_1^2 \frac{1}{x}\mathrm{d}x = \ln(2)$ Substitute $x = u^{-1} \rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = -u^{-2}$

$$= -\int_1^{\frac{1}{2}} \frac{u}{u^2}\mathrm{d}u = [\ln u]_{\frac{1}{2}}^{1} = \ln(2)$$

2D JACOBIAN

- For a continuous 1-to-1 transformation from (*x*,*y*) to (*u*,*v*)
- Then $x = x(u, v)$ and $y = y(u, v)$

- Where Region (in the *xy* plane) maps onto region R in the *uv* plane R'

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

2D Jacobian

maps areas dxdy to areas dudv

- Hereafter call such terms x_u

$$= \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = x_u y_v - x_v y_u$$

JACOBIAN IN 3D

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Suppose that x, y *and* z *are three independent variables which can be expressed in term of three other independent variables* u, v *and* w *by the formula*

$$x = x(u, v, w), y = y(u, v, w), z = z(u, v, w)$$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$$

PROPERTY OF JACOBIAN

- The Jacobian matrix $\dfrac{\partial(x,y)}{\partial(u,v)}$ is the ***inverse matrix*** of $\dfrac{\partial(u,v)}{\partial(x,y)}$ i.e.,

$$\begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- Because (and similarly for d*y)*

$$\mathrm{d}x = x_u \mathrm{d}u + x_v \mathrm{d}v = x_u \mathrm{d}u + x_v(v_x \mathrm{d}x + v_y \mathrm{d}y)$$

$$\begin{aligned} x \text{ constant } &\Rightarrow \mathrm{d}x = 0 \Rightarrow 0 = x_u u_y + x_v v_y \\ y \text{ constant } &\Rightarrow \mathrm{d}y = 0 \Rightarrow 1 = x_u u_x + x_v v_x \end{aligned}$$

- This makes sense because Jacobians measure the relative areas of d*x*d*y* and d*u*d*v*, i.e

$$\det(AB) = \det(A)\det(B) = 1 \Rightarrow \det(A) = \frac{1}{\det B}$$

- So

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \left|\frac{\partial(u,v)}{\partial(x,y)}\right|^{-1}$$

CHANGE OF VARIABLES

You will recall in one-dimensional calculus, when given an integral of the form $\int g'(u) f(g(u))\, du$, we performed the change of variable $x = g(u)$ which gave us $dx = g'(u)\, du$ and thus

$$\int g'(u) f(g(u))\, du = \int f(x)\, dx$$

We can write this slightly differently as follows. Since $x = g(u)$, $\dfrac{dx}{du} = g'(u)$ hence, we have

$$\int f(x)\, dx = \int f(g(u)) \frac{dx}{du} du$$

The Jacobian is what generalizes $\dfrac{dx}{du}$ in the above formula.

CHANGE OF VARIABLES

A change of variable is usually described as a transformation T from

uv-plane to the xy-plane given by $T(u,v) = (x,y)$
where x and y are given by

$$\begin{aligned} x &= x(u,v) \\ y &= y(u,v) \end{aligned}$$

We assume that the first order partials of $x(u,v)$ and $y(u,v)$ are continuous. When it is the case we say that T is a C^1 **transformation**. Such a transformation will map a region S in the uv-plane into another region R into the xy-plane

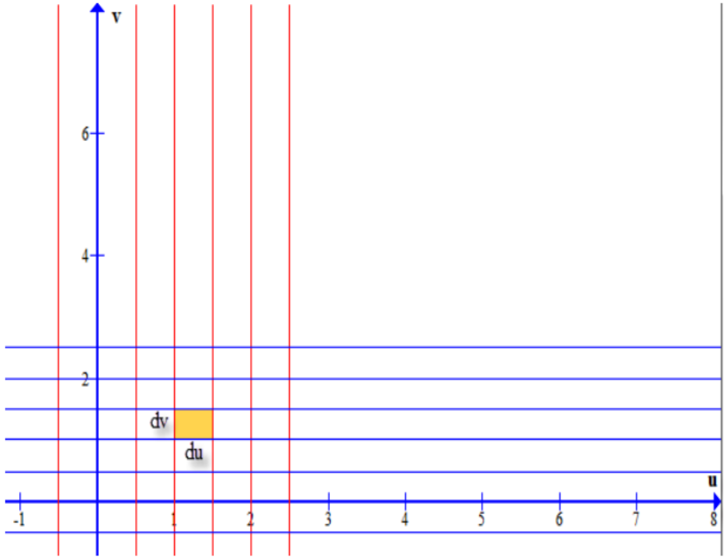
In double integral, we divided the region of integration into small regions.

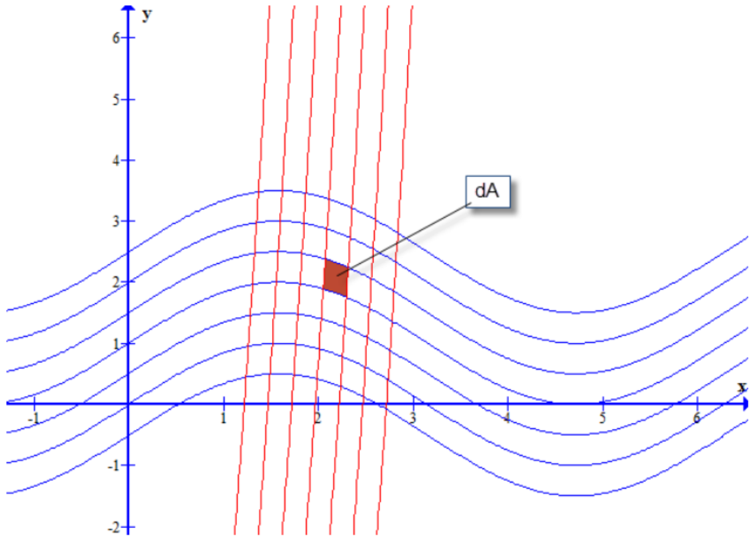
The region of integration was then approximated by rectangles of area

$$dA = dudv.$$

When we apply the transformation T,
uv-plane and the grid are transformed as well as the region of integration.

TRANSFORMATION FROM XY PLANE TO UV PLANE





small region created by the grid are no longer rectangles
their area dA is not $dxdy$.

INTERPRETATION OF JACOBIAN

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From the differential formula, we see that if $x = x(u, v)$ then

$$dx = \frac{\partial x}{\partial u}du + \frac{\partial x}{\partial v}dv$$

Similarly, if $y = y(u, v)$ then

$$dy = \frac{\partial y}{\partial u}du + \frac{\partial y}{\partial v}dv$$

We can write this in matrix form as follows:

$$\begin{bmatrix} dx \\ dy \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \begin{bmatrix} du \\ dv \end{bmatrix}$$

THANK YOU